

Assembly Guide

LoFi Generative Sequencer

Regional Electronic Music Australia



The LoFi Generative Sequencer is a self-contained, DIY electronic musical instrument. It is battery powered, with a built-in loudspeaker and headphone/line output, and driven by an ESP32 microcontroller. It can generate and playback repeated monophonic musical patterns up to 16 steps in length.

Parts List:

- ESP32-C3 supermini, x1
- Buttons, Green & Blue
- Potentiometers, x4
- Pot Caps, White & Red & Yellow & Green
- LEDs, Red & Green
- Switch x1
- Audio Jack 6.35mm x1

- Loudspeaker 100 mm x1
- Breadboard x1
- Jumper Wire (short x3, long x2)
- Resistors, 220 Ohm x 2 (LEDs)
- Capacitor, 0.1uF 104 x1 (audio)
- Resistor, 10K Ohm x1 (audio)
- Battery Case x1
- Batteries AA x 3
- Amplifier board and screw terminal x1
- Enclosure panels, top & bottom & back & front
- Rubber Feet x4
- Standoffs 10mm x2 (speaker)
- Standoffs with sockets 30mm x2
- Standoffs with thread 30mm x2
- Bolts for standoffs M3 6mm x6
- Washers for speaker standoffs x4 (two per 10mm standoff)

Electronic Parts and Wire Colours:

- The green Button has two wires, Pin (xxx) and Ground (xxx), green and yellow
- The blue Button has two wires, Pin (xxx) and Ground (xxx)
- Potentiometers have three wires, Power (xxx), Pin (xxx), Ground (xxx)
- LEDs have two wires, Ground (xxx) and Pin/Power via a resistor (xxx)
- Switch has two wires (one leg is unconnected), Power source (xxx), Power destination (xxx).
- The Loudspeaker has two wires, positive (xxx) and negative (xxx).
- The Amp Board has five header pins, A+ (audio), A- (Ground), SD (unconnected), Vcc (Power), Gnd (Ground). Also, two screw terminals for loudspeaker + and -.
- The ESP32 C3 supermini has sixteen header pins, three for power (in/out) and ground, and the rest are General Purpose Input Output (GPIO) pins. Many of the GPIOs labelled 0 – 10 are used for this project.

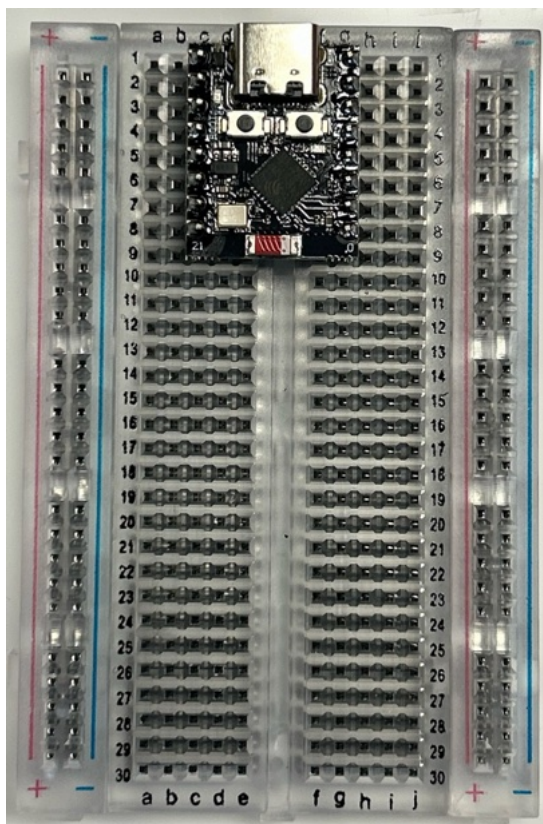
An overview of the construction:

1. The LoFi Sequencer is built on a 'breadboard' using pre-soldered components. The ESP32 and Amplifier are mounted on the breadboard via header pins.
2. These components are connected to power and ground with jumper wires
3. Three resistors and a capacitor are added to the breadboard. A 10K resistor and 0.1uF capacitor for audio filtering, and two 220 Ohm resistors for LED power management.
4. Buttons and dials are attached to the top wooden plate.
5. The loudspeaker is connected to the top wooden plate via two 10mm offsets.

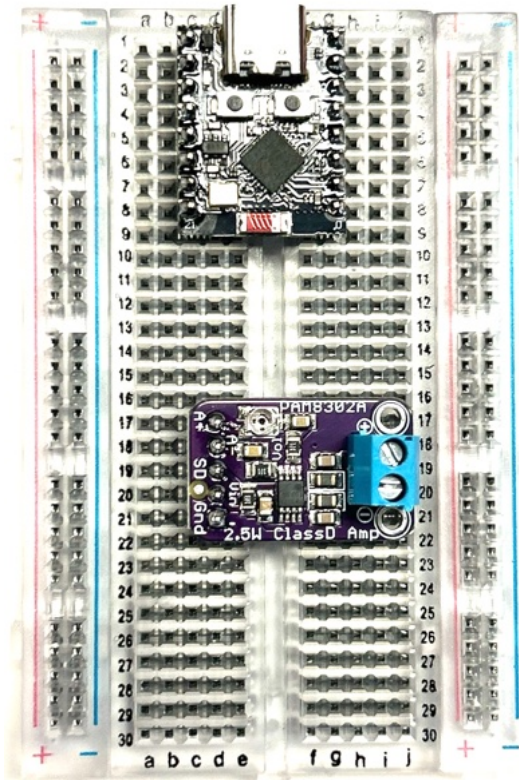
6. The audio socket is screwed into the wooden back side panel.
7. Wires from the buttons, dials, switch, speaker, jack socket, and battery pack are connected to the breadboard.
8. Test that everything works as expected before proceeding further.
9. The top and bottom panels are secured to each other with offsets screwed to the plates.
10. The breadboard (peel off sticker), battery pack (hot glue) are affixed to the wooden bottom laser cut panels.
11. Retest that the device works as expected before proceeding further.
12. Apply small amounts of hot glue to the 'dupont' connections on the breadboard to make them secure.
13. Attach the back and front panels to the top and bottom plates to complete the enclosure.

Detailed construction steps:

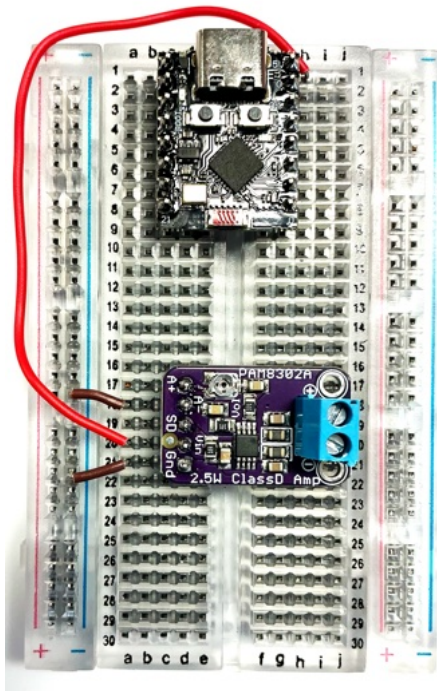
Add the chips to the board



1. Place the ESP32 chip on the top of the breadboard



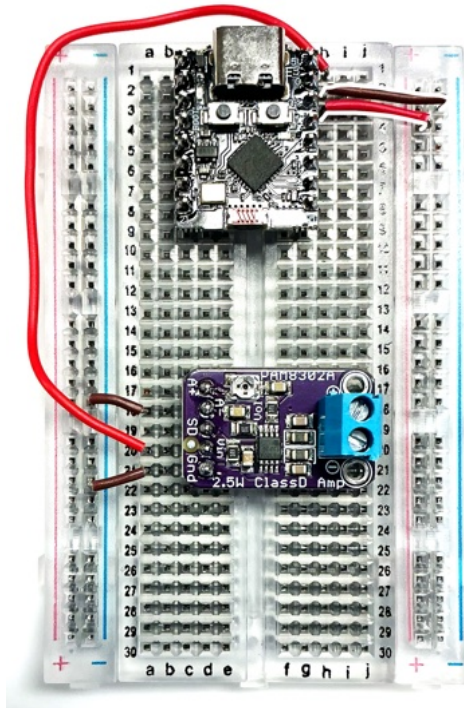
2. Place the amplifier chip on the breadboard at row 17



3. Add the power connectors to the amp board

Pin 2 of the amplifier board (in row 18 on the breadboard) goes to ground.

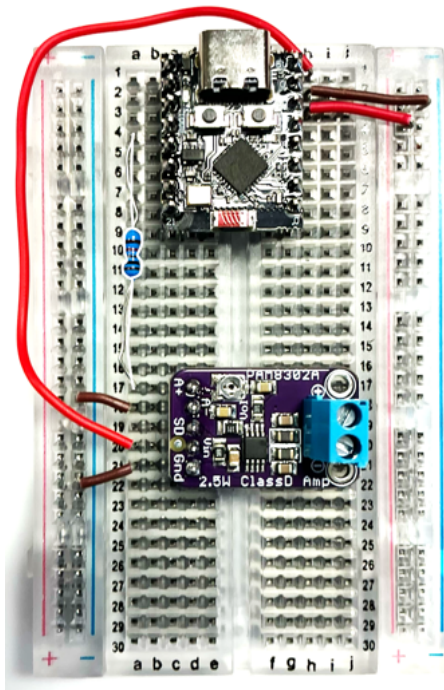
Pin 4 (row 20) of the amplifier board connects to pin 9 of the ESP32 (the top right pin), pin 5 of the amplifier board (labelled gnd in row 21) goes to ground.



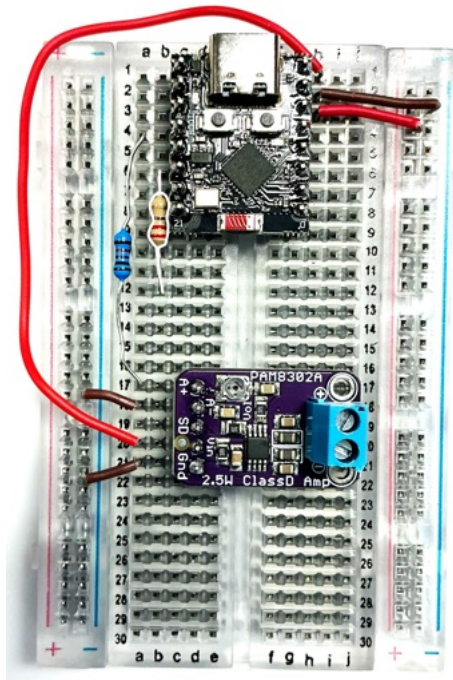
4. Add power connectors to the ESP32.

Pin 10, which is the second pin down on the right hand side goes to ground.

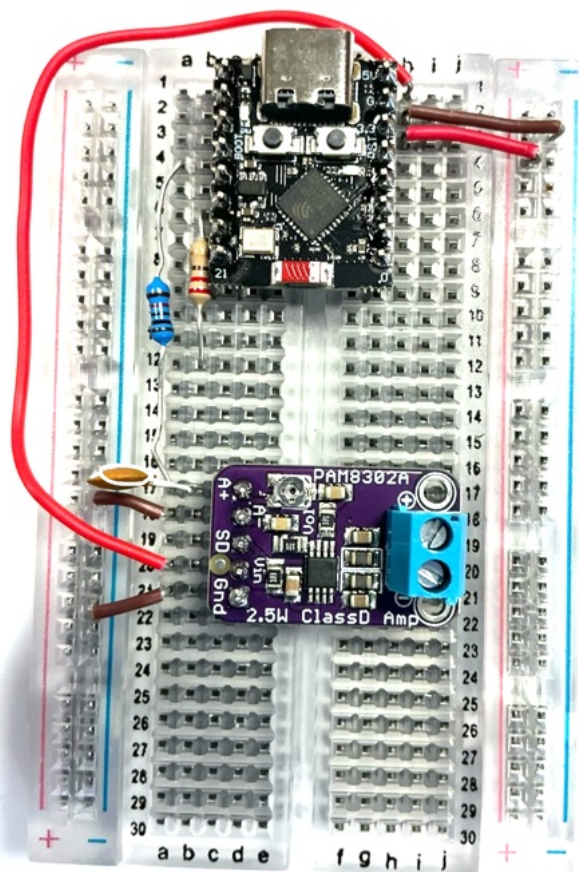
Pin 11, which is the third pin down on the right hand side goes to positive power.



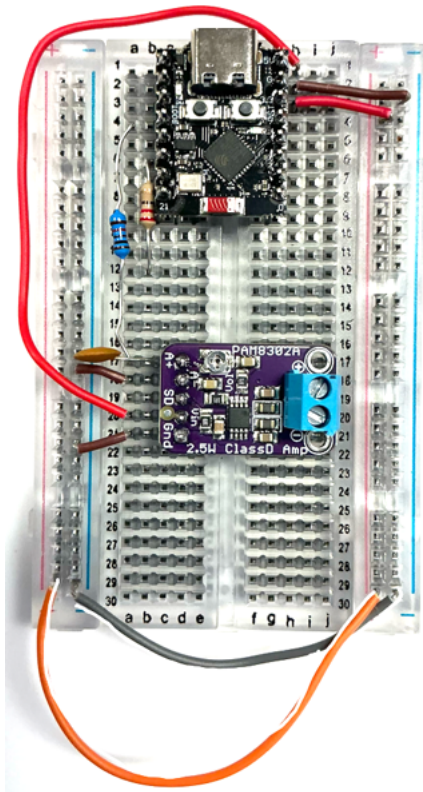
5. Put a 10k resistor (the blue one) between pin 4 on the ESP32 and pin 1 (A+) of the amplifier.



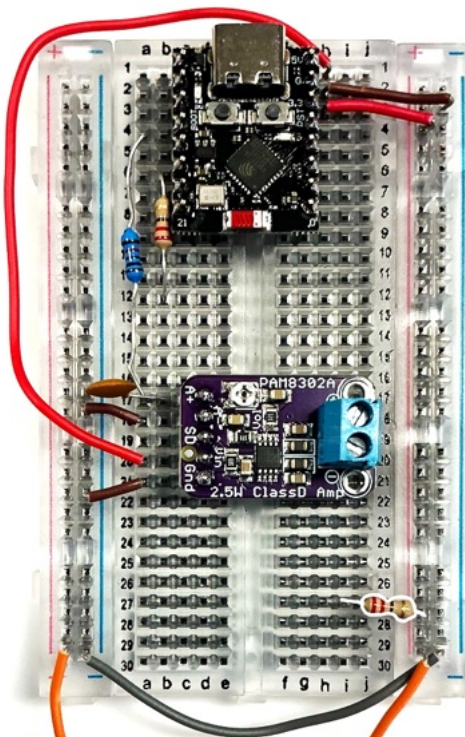
6. Add a 220 ohm resistor (the yellow one) between pin 6 of the ESP32 and an empty row. I used row 12. This is for our power led. Cut the legs a little so that it sits flush to the board. Make sure the two resistors are not touching.



7. Add the capacitor between pin 1 of the amplifier chip, labelled A+ in row 17 and the ground rail. Cut the legs a little so that it sits low to the board.



8. Connect the board power rails together.
Make sure the positive connects to positive and ground to ground.



9. Add a 220 ohm resistor in row 27 to the power rail.
This for our power led that we will add later

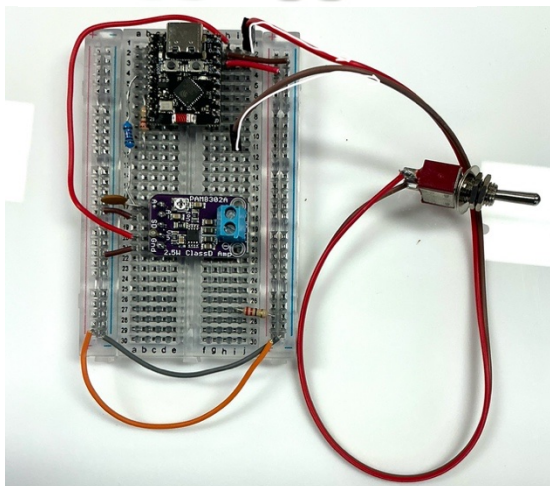
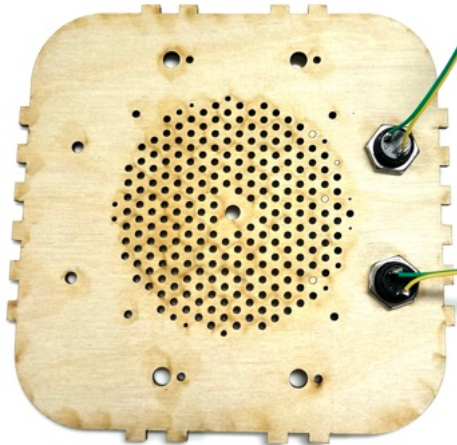
External components.

These components are attached to the box.

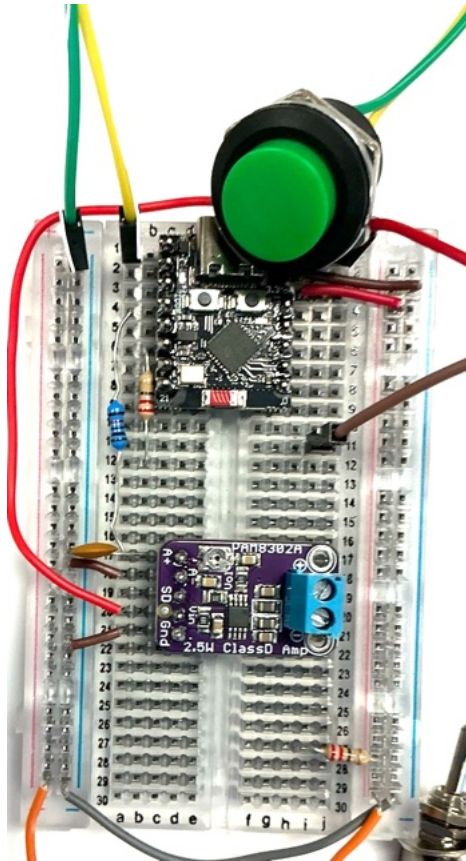
Some external components, such as the buttons, should be attached to the lid before being connected to the bread board.



10. Screw the buttons into the lid of the sequencer. The green button goes on the left.

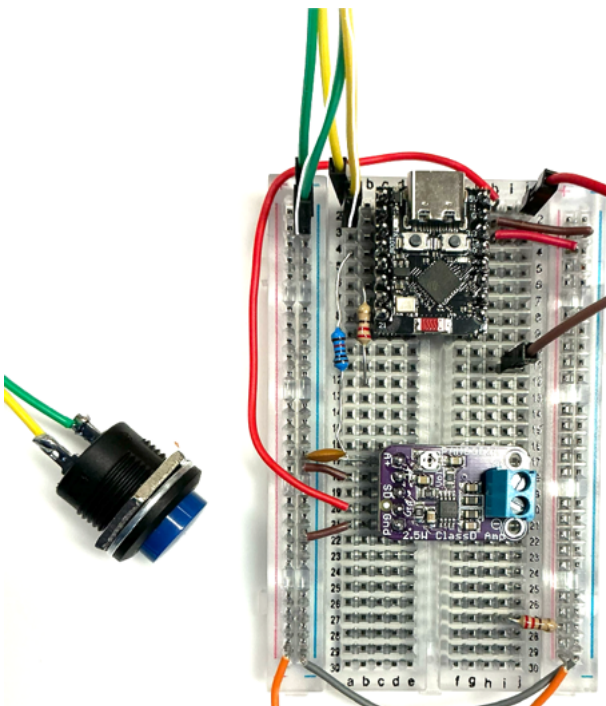


11. Add the switch to the breadboard. one side goes to pin 9 of the ESP32 *take note - that is the top pin on the right hand side of the chip*. The other side goes to an empty breadboard row – eg row 11 on the right



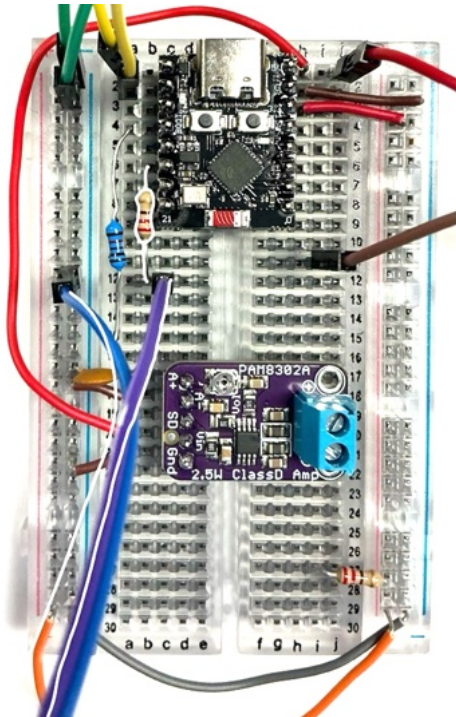
12. Add the green button.
One side of it connects to pin 1 of the ESP32 the other side into ground. It doesn't matter which way around this goes.

* Your button should be attached to the sequencer of lid, unlike in this picture.



13. Add the blue button
between pin 2 of the ESP32 and ground. It doesn't matter which way around this goes.

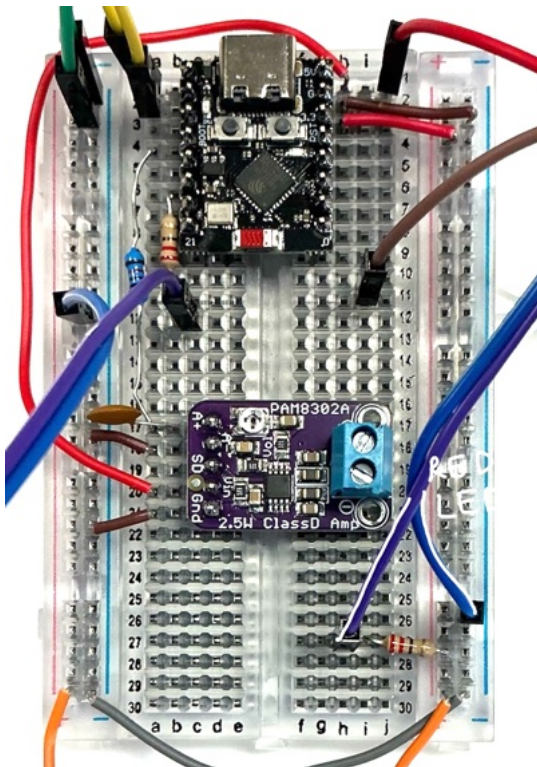
* Your button should be attached to the sequencer of lid, unlike in this picture



14. Add the green LED.

The blue cable goes to ground.

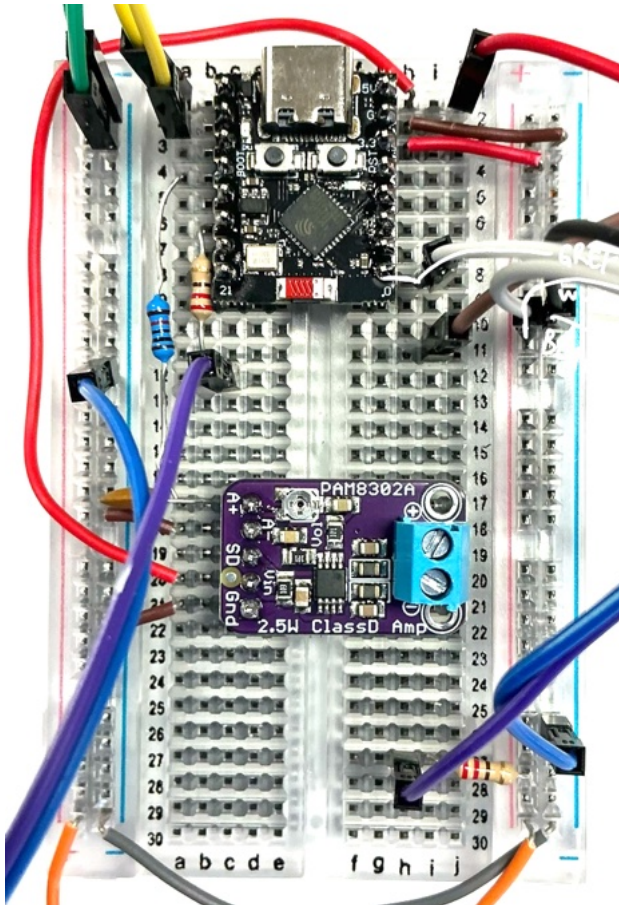
The purple cable goes to the resistor in row 12.



15. Add the red LED.

The blue cable goes to ground

The purple cable goes to the resistor in row 27.



16. Add the potentiometers.

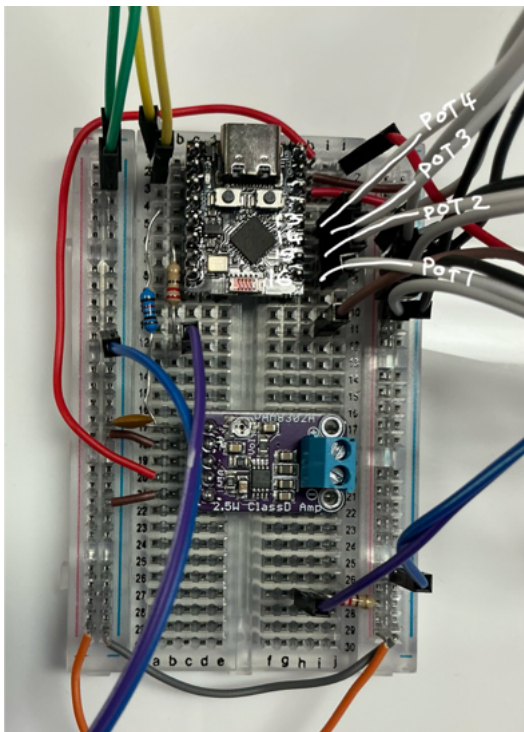
Each potentiometer has three leads.

The black leads go to ground,

The white leads go to the positive rail.

The grey ones will go to a specific pin.

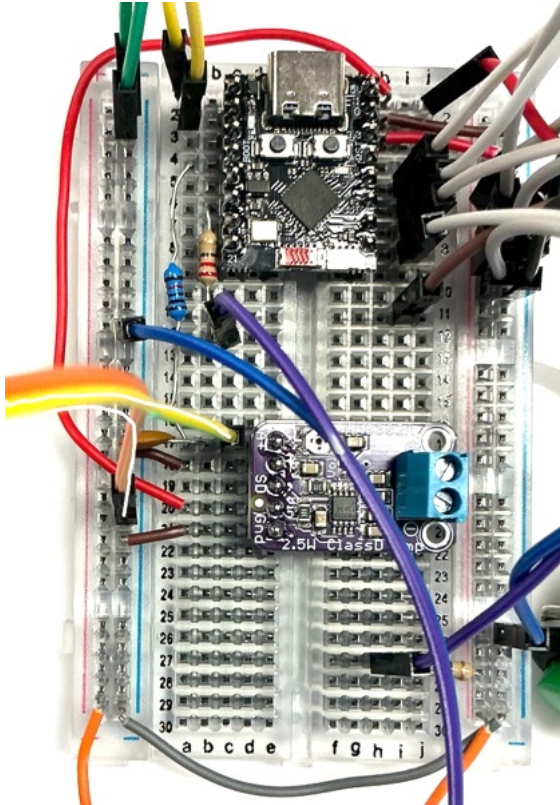
Potentiometer 1 grey lead goes to pin 16 on the ESP32



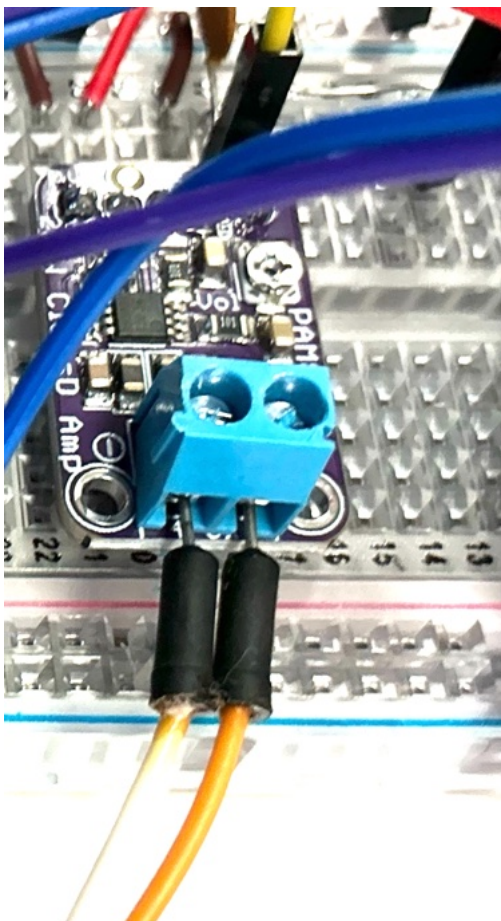
Potentiometer 2 grey lead goes to pin 15 on the ESP32

Potentiometer 3 grey lead goes to pin 14 on the ESP32

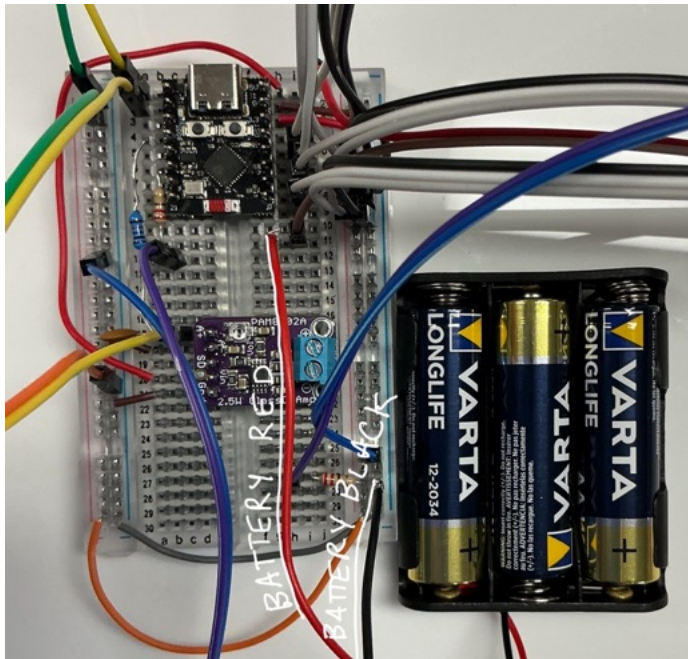
Potentiometer 4 grey lead goes to pin 13 on the ESP32



17. Add the output jack.
Connect the orange lead to ground and the yellow lead to pin 1 (marked A+) on the amplifier board



18. Connect the speaker to the screw terminals



19. Connect battery.

The red cable goes to the switch in row 11 of the breadboard.

The black cable goes to ground.

NOW LET'S TEST THAT IT WORKS!

Plug the batteries in and you should hear some sound and the lights will turn on.

Glue! If everything is working now, it's time to get the hot glue and use it to keep everything in place permanently.

Important! Your buttons need to already be screwed onto the lid. If they are not pull them off the bread board screw them onto the lid and then place them back into the bread board and check everything is working again.

BOX ASSEMBLY

Once everything is working and glued down in place, we can now screw the rest of the components onto the box. Your buttons should already be screwed on.



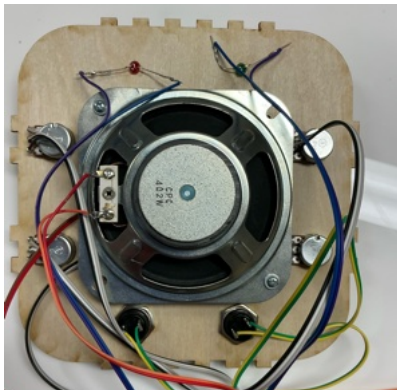
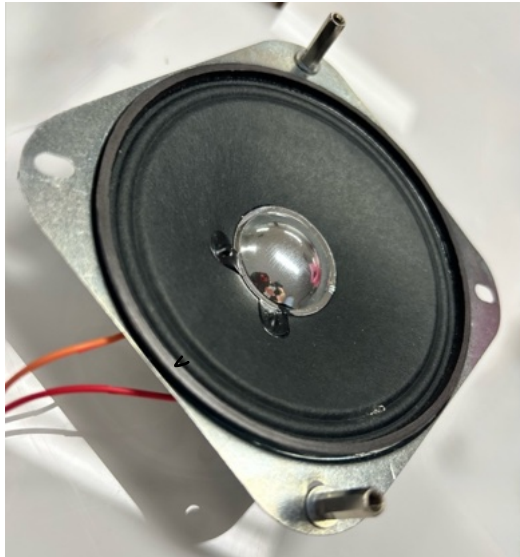
20. Screw the switch into the centre hole on the lid. The nut and bolt go on the top.



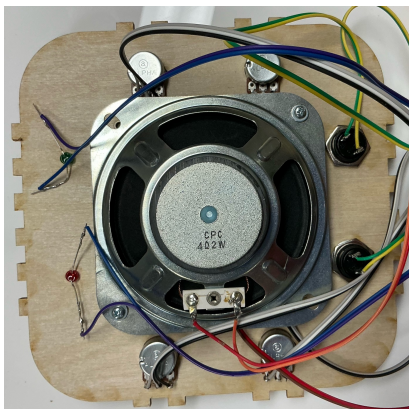
21. Attach the speaker to the lid.

Two diagonally opposite corners will attach to the lid with a small stand-off.

Use a small washer on each side of the speaker hole.



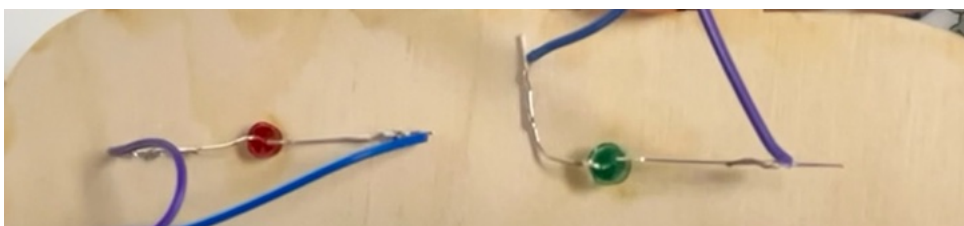
22. Attach the potentiometers to the lid



23. Poke the LEDs through the hole

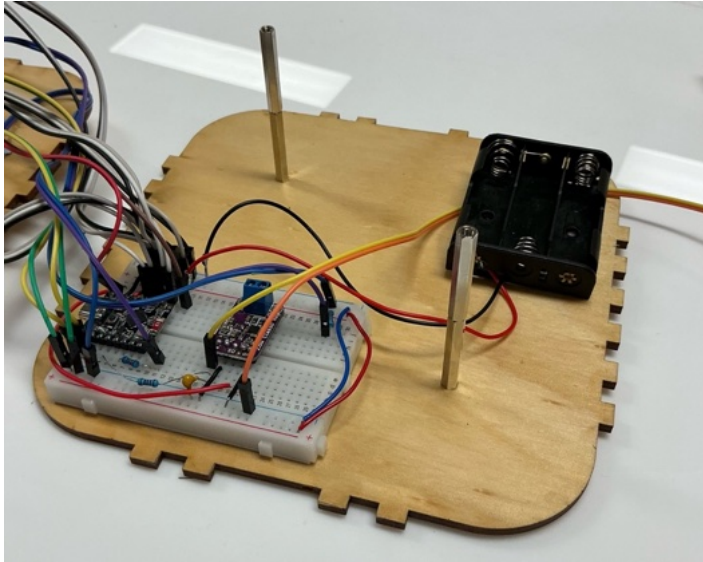
Flatten the legs so they cannot touch each other and cause a short circuit.

Use hot glue to stick them down, gluing the legs in place.

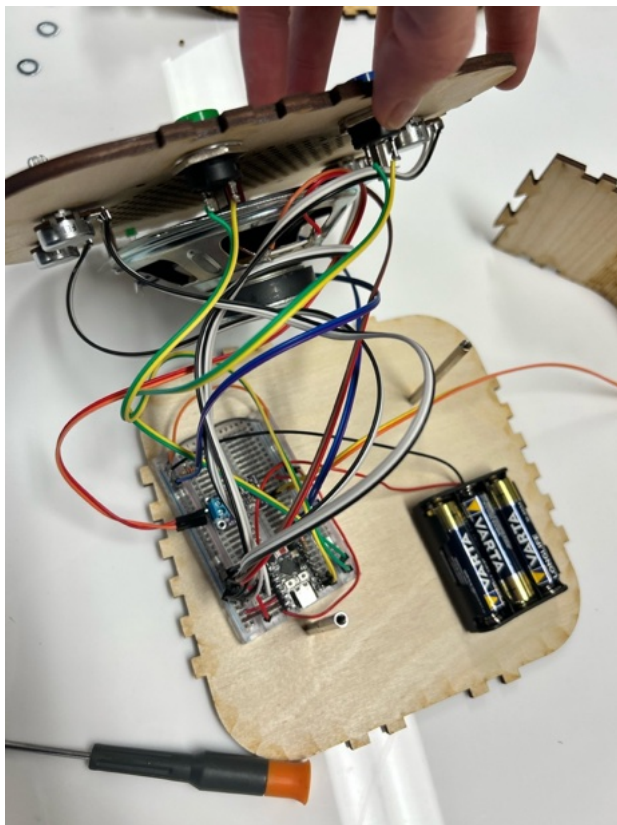




Connect the two long stand-offs together x 2

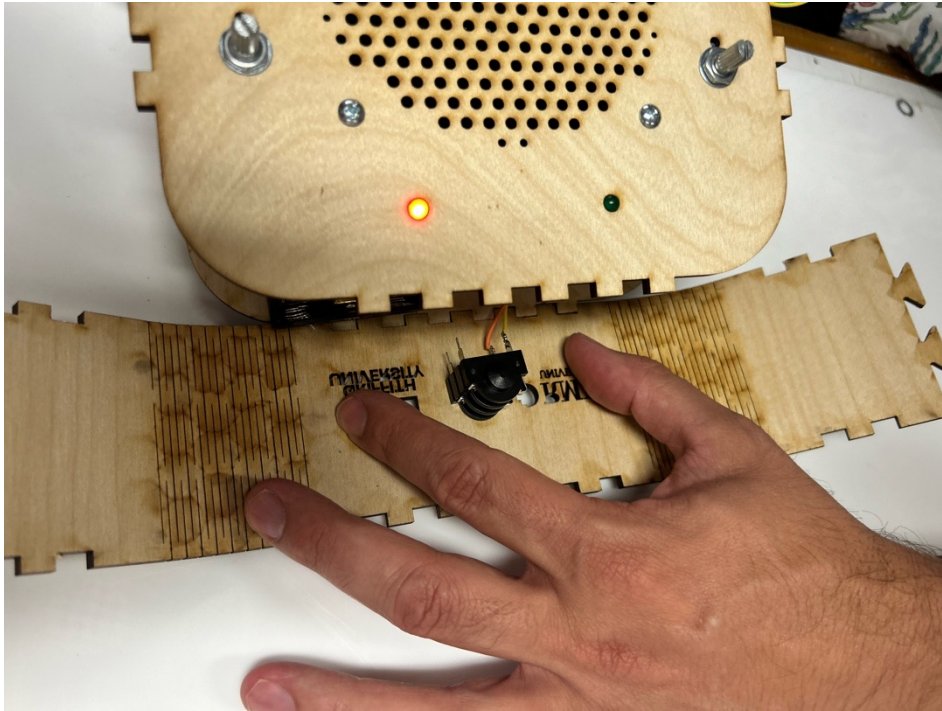


Screw the Connected stand-offs to the sequencer base. Check the positioning of the screws of the speaker on the lid. And make sure the stand-offs will be on the opposite corners.

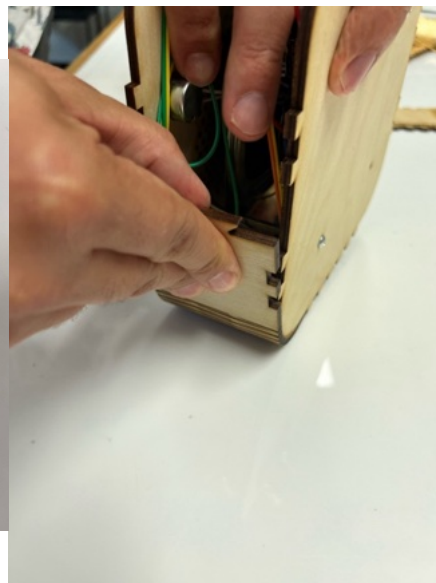
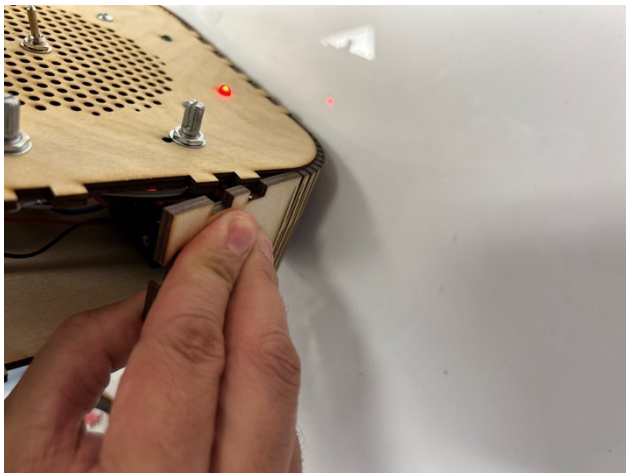


Place and glue down the bread board and the battery pack. Make sure they sit in a space that gives room for the speaker.

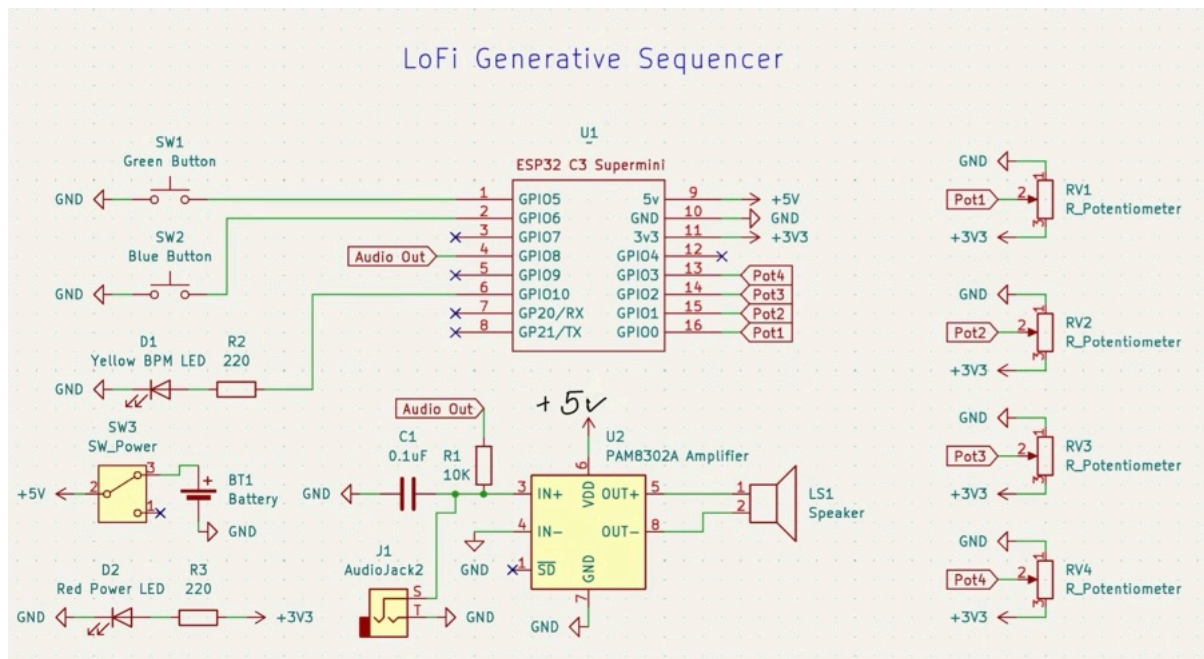
Put it all together



1. Lineup the teeth of the side with the jack on it
2. Then connect the teeth on the top side.
3. Place it on its side and gently stretch the wooden hinges to close the side teeth.
4. Repeat this with the other side.



Schematic:



Files:

All files required for building the LoFi Sequencer are available on the device's [GitHub](https://github.com/algomusic/lofi-sequencer) site - <https://github.com/algomusic/lofi-sequencer>